InnovateX 2.0

IEEE EMBS
Pune Chapter

Loan Default Risk Prediction

from Pre-Loan Transaction Behavior

- Problem Statement ID - Internal project
- Problem Statement Title - Loan Default Risk Prediction
- Theme - Data Mining and Predictive Analytics
- PS Category - Software
- Team ID - To be filled
- Team Name - To be filled

PROBLEM STATEMENT

Banks lose money when loans default, even though useful warning signals may already exist in customer transaction history before the loan is granted.

- The project predicts whether a loan will be repaid or default using only pre-loan transaction behavior.
- It uses 682 loans from the PKDD'99 Czech bank dataset: 606 repaid and 76 defaulted (11.1% default rate).
- The solution supports earlier, evidence-based risk screening and helps reduce avoidable credit losses.

OBJECTIVE

- Build a complete data-mining pipeline for loan default risk prediction.
- Engineer 13 leakage-checked behavioral features from each account's history before the loan date.
- Compare six classifiers with and without SMOTE, tune their hyperparameters, and select the most reliable model using repeated cross-validation.
- Translate model findings into practical risk indicators and customer behavior segments.

SCOPE

TARGET USERS

- Credit analysts
- Risk managers
- Banking decision-makers

DATA AND APPLICATION

- PKDD'99 financial dataset
- Pre-loan account activity
- Loan-level risk screening

KEY FEATURES

- Preprocessing and EDA
- Classification and tuning
- Segmentation and dashboards

The prototype is designed for analytical decision support; final lending decisions remain subject to banking policy and human review.

PROPOSED SOLUTION

A reproducible Python pipeline converts raw customer records into an interpretable loan-risk assessment.

01
Raw data
loan, transaction, account, client, disp

>

02
Features
13 pre-loan behavior measures

>

03
Models
6 classifiers + tuning + SMOTE

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04
Decision support
risk scores, patterns, segments

Selected model: tuned Gradient Boosting (repeated-CV F1 0.571 +/- 0.120, Average Precision 0.654, ROC-AUC 0.864). Tuned Random Forest forms the leading group with ROC-AUC 0.877.

TECHNOLOGY STACK

LANGUAGE
Python 3.13.5

DATA
pandas, NumPy

ML
scikit-learn, imbalanced-learn

VISUALIZATION
Matplotlib, Seaborn

WORKFLOW
Jupyter, nbconvert

OUTPUT
CSV, PNG, PKL, dashboard

Training-only preprocessing pipelines protect against leakage, while repeated stratified cross-validation supports honest model selection under class imbalance.

INNOVATION

- Uses transaction behavior that already exists before loan approval, rather than relying only on static demographics.
- Combines predictive classification with interpretable feature importance, risk quadrants, association patterns, and KMeans customer segmentation.
- Evaluates the honest minority-class metrics: F1, Average Precision, and ROC-AUC, instead of accuracy alone.
- Finds that low pre-loan balances and balance at loan time are stronger risk signals than age or gender, enabling focused and explainable screening.

Added value

A practical bridge between data mining coursework and a bank-ready analytical workflow: reproducible, measurable, and designed to support human decisions.